Generative AI Questions for Session I

UNIT I

1. Differentiate between Generative and Discriminative models with suitable examples.
2. Explain the architecture and working of GANs with Generator, Discriminator, training process, and challenges.
3. Describe the architecture of Variational Autoencoders (VAEs) and explain the significance of the Reparameterization.
4. Discuss the concept of Diffusion Models and explain the role of Vector Quantization (VQ) in latent diffusion.
5. Discuss the ethical aspects of Generative AI and explain the principles of Responsible AI.
6. A VAE is trained on handwritten digit images. Explain how the model can generate a new digit that was not present exactly in the training dataset.
7. A company wants to automatically generate product descriptions from product information. Explain how a Generative AI system could be designed for this application

UNIT II

1. Draw the block diagram of the Transformer model and explain the role of the Encoder and Decoder blocks.
2. Compare and contrast BERT (Bidirectional Encoder Representations from Transformers) and GPT (Generative Pre-trained Transformer).
3. What is Prompt Engineering? Explain Zero-Shot, Few-Shot, and Chain-of-Thought (CoT) prompting techniques with examples.
4. What is Hallucination in Large Language Models, and how can it be mitigated?
5. Explain RLHF and describe its major stages in training an LLM.
6. A student wants an LLM to summarize a research paper in exactly five points. Design an appropriate prompt with suitable constraints.
7. Explain the complete workflow of a RAG system with simple example